

E-Commerce Inventory & Sales Forecasting

A Project Report for Internship

Submitted by

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In partial Fufillment for the award of the degree of

B.TECH CSE-AIML

MCKV INSTITUTE OF ENGINEERING



At





BONAFIDE CERTIFICATE

Certified that this project work was carried out under my supervision

“ **E-Commerce Inventory & Sales Forecasting** ” is the Bonafide work of

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Acknowledgement

I would like to express my sincere gratitude to my project mentor, **Mr. Raihan Mistry**, and the faculty members for their valuable guidance and support throughout the development of this project.

I am thankful for the opportunity to work on the “**E-Commerce Inventory & Sales Forecasting**” project, which helped me understand the practical applications of data analytics, SQL, Python, and Power BI.

I would also like to express my sincere gratitude to **Euphoria Gen x** for their unwavering support and encouragement throughout the project.

This project helped me improve my practical knowledge of **data cleaning, exploratory data analysis, SQL queries, demand forecasting, and interactive dashboard development.**

Data Model

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	
1	Date	Store ID	Product ID	Category	Region	Inventory L	Units Sold	Units Orde	Demand F	Price	Discount	Weather	C Holiday/Pr	Competito	Seasonalit	Year	Month	Month_Nai	Day	Day_of_We	Week	Revenue		
2	01-01-2022	S001	P0001	Groceries	North	231	127	55	135.47	33.5	20	Rainy		0	29.69	Autumn	2022	1	January	1	5	52	4254.5	
3	02-01-2022	S001	P0001	Groceries	West	116	81	104	92.94	27.95	10	Cloudy		0	30.89	Spring	2022	1	January	2	6	52	2263.95	
4	03-01-2022	S001	P0001	Electronic	West	154	5	189	5.36	62.7	20	Rainy		0	58.22	Winter	2022	1	January	3	0	1	313.5	
5	04-01-2022	S001	P0001	Groceries	South	85	58	193	52.87	77.88	15	Cloudy		1	75.99	Winter	2022	1	January	4	1	1	4517.04	
6	05-01-2022	S001	P0001	Groceries	South	238	147	37	150.27	28.46	20	Sunny		1	29.4	Winter	2022	1	January	5	2	1	4183.62	
7	06-01-2022	S001	P0001	Electronic	East	198	37	155	39.09	91.05	5	Snowy		0	87.3	Summer	2022	1	January	6	3	1	3368.85	
8	07-01-2022	S001	P0001	Furniture	North	195	107	106	117.92	53.99	10	Sunny		1	58.8	Winter	2022	1	January	7	4	1	5776.93	
9	08-01-2022	S001	P0001	Furniture	East	231	2	119	0.84	66.3	20	Cloudy		1	63.82	Winter	2022	1	January	8	5	1	132.6	
10	09-01-2022	S001	P0001	Electronic	South	373	350	178	352.24	41.72	0	Sunny		1	38.68	Summer	2022	1	January	9	6	1	14602	
11	10-01-2022	S001	P0001	Electronic	West	327	36	132	39.5	32.68	5	Snowy		1	34.71	Summer	2022	1	January	10	0	2	1176.48	
12	11-01-2022	S001	P0001	Furniture	West	101	16	193	16.11	27.53	5	Rainy		1	27.08	Autumn	2022	1	January	11	1	2	440.48	
13	12-01-2022	S001	P0001	Toys	South	145	104	189	100.32	92.92	0	Snowy		0	97.59	Winter	2022	1	January	12	2	2	9663.68	
14	13-01-2022	S001	P0001	Furniture	East	110	80	190	78.64	17.27	15	Snowy		1	21.79	Spring	2022	1	January	13	3	2	1381.6	
15	14-01-2022	S001	P0001	Toys	North	199	177	44	193.23	90.05	20	Snowy		0	87.05	Winter	2022	1	January	14	4	2	15938.85	
16	15-01-2022	S001	P0001	Groceries	North	290	176	94	170.06	98.04	10	Rainy		1	98.2	Spring	2022	1	January	15	5	2	17255.04	
17	16-01-2022	S001	P0001	Groceries	East	104	97	189	114.85	11.39	15	Snowy		0	7.96	Winter	2022	1	January	16	6	2	1104.83	
18	17-01-2022	S001	P0001	Groceries	East	134	69	95	73.59	95.27	0	Rainy		1	92.6	Summer	2022	1	January	17	0	3	6573.63	
19	18-01-2022	S001	P0001	Electronic	North	339	123	157	134	90.7	15	Sunny		0	86.51	Spring	2022	1	January	18	1	3	11156.1	
20	19-01-2022	S001	P0001	Groceries	West	431	211	139	228.92	97.97	10	Cloudy		0	95.06	Summer	2022	1	January	19	2	3	20671.67	
21	20-01-2022	S001	P0001	Furniture	North	154	79	113	90.49	10.38	0	Cloudy		0	11.46	Spring	2022	1	January	20	3	3	820.02	
22	21-01-2022	S001	P0001	Clothing	South	430	390	195	401.1	13.1	15	Snowy		1	15.14	Summer	2022	1	January	21	4	3	5109	
23	22-01-2022	S001	P0001	Electronic	North	450	191	197	181.42	87.64	0	Snowy		1	91.45	Spring	2022	1	January	22	5	3	16739.24	
24	23-01-2022	S001	P0001	Furniture	West	118	79	155	69.99	22.65	15	Cloudy		0	25.19	Spring	2022	1	January	23	6	3	1789.35	
25	24-01-2022	S001	P0001	Toys	South	115	9	132	17.87	76.71	15	Snowy		1	75.85	Summer	2022	1	January	24	0	4	690.39	
26	25-01-2022	S001	P0001	Furniture	East	348	189	122	185.83	18.35	15	Rainy		1	22.01	Autumn	2022	1	January	25	1	4	3468.15	
27	26-01-2022	S001	P0001	Clothing	South	200	49	101	50.62	59.48	5	Sunny		1	55.45	Winter	2022	1	January	26	2	4	2914.52	
28	27-01-2022	S001	P0001	Furniture	South	410	164	116	180.45	93.34	0	Cloudy		0	94.33	Winter	2022	1	January	27	3	4	15307.76	
29	28-01-2022	S001	P0001	Furniture	East	158	97	82	115.96	53.25	20	Cloudy		1	49.95	Autumn	2022	1	January	28	4	4	5165.25	
30	29-01-2022	S001	P0001	Furniture	South	154	47	124	48.79	88.06	15	Snowy		1	92.84	Autumn	2022	1	January	29	5	4	4138.82	
31	30-01-2022	S001	P0001	Furniture	West	498	130	63	148.26	68.76	10	Snowy		0	73.29	Spring	2022	1	January	30	6	4	8938.8	

- The dataset consists of historical retail sales and inventory records covering multiple stores, products, categories, regions and time periods.
- The main columns include Date, Store ID, Product ID, Category, Region, Inventory Level, Units Sold, Units Ordered, Demand Forecast, Price, Discount, Seasonality and Revenue.
- Data cleaning and preparation were carried out using Python (Pandas). Additional date-related columns and revenue were included in the cleaned dataset.
- ARIMA forecasting was used to generate product-level demand predictions, and the results were saved in a separate forecast output file.
- MySQL was used to create the retail sales database and perform revenue, sales, category, regional, monthly and product-level analysis. A category_performance view was also created.
- Power BI was used to develop five dashboard views covering Executive Overview, Sales Performance, Inventory Monitoring, Product & Regional Analysis, and the overall dashboard.

Overview Page



The Executive Overview page presents:

- **Total Revenue** – shows the overall revenue generated.
- **Total Units Sold** – shows the total number of units sold.
- **Total Inventory** – shows the total inventory available.
- **Average Demand Forecast** – shows the average forecasted demand.
- **Total Products** – shows the number of products included in the analysis.
- **Total Stores** – shows the number of stores included in the dataset.
- **Revenue by Category** – compares revenue across different product categories.
- **Revenue by Region** – compares revenue across different regions.
- **Monthly Revenue Trend** – shows how revenue changes month by month.
- **Region Filter** – allows the user to select a specific region.
- **Category Filter** – allows the user to select a specific product category.
- **Seasonality Filter** – allows the user to analyze the data based on seasonality.

Sales Performance Dashboard.



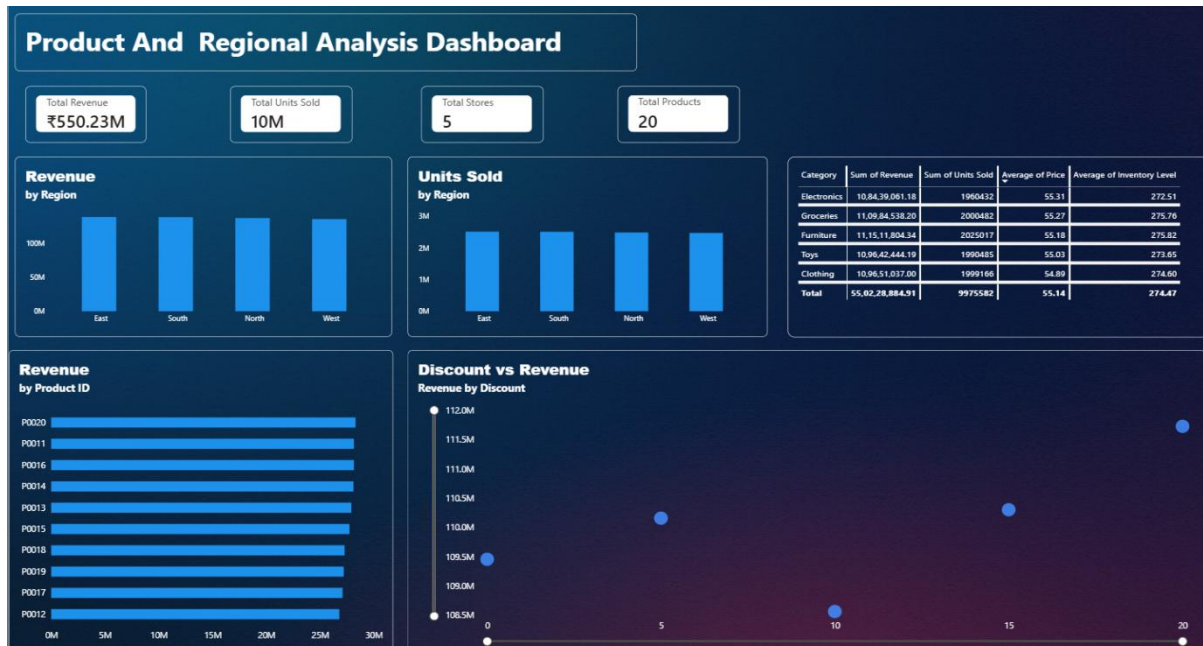
- **Total Revenue** – shows the overall revenue generated from sales.
- **Total Units Sold** – shows the total number of units sold.
- **Average Price** – shows the average selling price of the products.
- **Average Demand Forecast** – shows the average forecasted demand.
- **Revenue by Category** – compares revenue generated by different product categories.
- **Units Sold by Category** – compares the number of units sold across categories.
- **Top 10 Products by Revenue** – displays the products with the highest revenue contribution.
- **Monthly Revenue Trend** – shows the change in revenue across different months.
- **Category and product-level analysis** helps in understanding sales performance in more detail.

INVENTORY MONITORING



- **Total Inventory** shows the overall inventory available in the dataset.
- **Total Units Ordered** shows the number of units ordered.
- **Average Inventory Level** provides the average inventory level across the records.
- **Average Demand Forecast** shows the expected demand based on the forecasting output.
- **Inventory Level vs Demand Forecast** compares current inventory with forecasted demand.
- **Units Ordered by Category** compares ordered units across different categories.
- **Inventory by Category** shows inventory distribution across product categories.
- **Product Inventory Details** provides product-level information including inventory, units sold, units ordered and demand forecast.
- This dashboard helps understand the relationship between **inventory levels and expected demand**.

PRODUCT & REGIONAL ANALYSIS



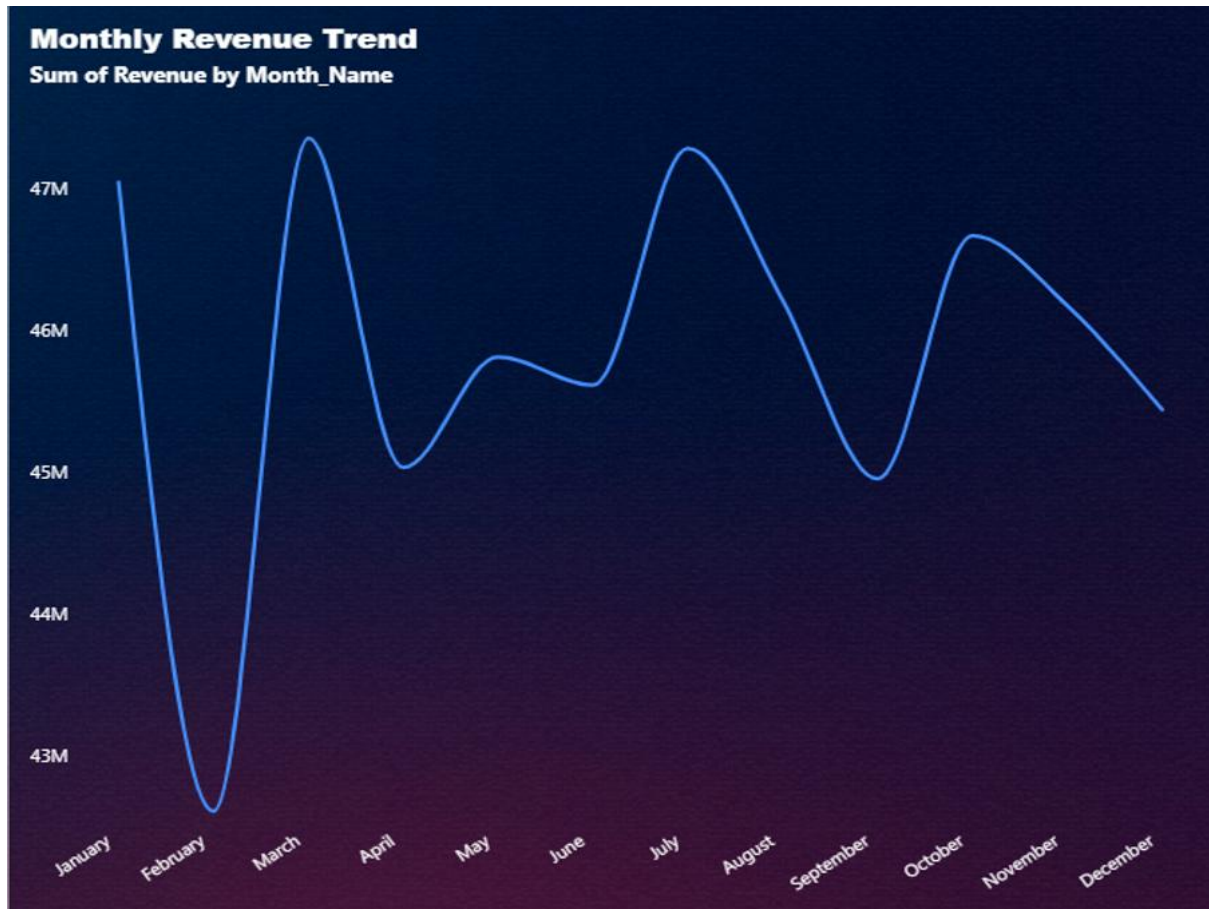
- The dashboard provides an analysis of **product and regional performance**.
- **Revenue by Region** compares revenue generated across different regions.
- **Units Sold by Region** compares sales volume across regions.
- **Revenue by Product** shows the revenue contribution of individual products.
- **Category Performance** provides a comparison of revenue, units sold, average price and inventory.
- **Discount vs Revenue** provides a visual comparison between discount levels and revenue.
- The dashboard helps identify differences in **product, category and regional performance**.

DEMAND FORECASTING

Store ID	Product ID	Category	Region	Forecast Date	Forecasted Demand	Forecast Meth	Inventory Level	Average Daily Demand
S001	P0001	Toys	East	01-02-2024	949.1931790962058	ARIMA	223	137.3064295485636
S001	P0002	Furniture	West	01-02-2024	672.7658414700047	ARIMA	217	131.03830369357044
S001	P0003	Toys	East	01-02-2024	3201.169958529687	ARIMA	69	141.4404924760602
S001	P0004	Groceries	West	01-02-2024	4567.401493429397	ARIMA	338	140.90424076607388
S001	P0005	Groceries	North	01-02-2024	5272.718052315378	ARIMA	471	133.1299589603283
S001	P0006	Electronic	North	01-02-2024	2355	ARIMA + 3-Mor	305	130.6730506155951
S001	P0007	Clothing	East	01-02-2024	4707.778054596665	ARIMA	256	137.8235294117647
S001	P0008	Electronic	East	01-02-2024	1345.1952857776114	ARIMA	315	137.1326949384405
S001	P0009	Electronic	South	01-02-2024	4223.275700707657	ARIMA	167	131.43638850889192
S001	P0010	Furniture	West	01-02-2024	2151.578012153738	ARIMA	167	132.3611491108071
S001	P0011	Electronic	West	01-02-2024	2991.724765	ARIMA	449	132.4924760601915
S001	P0012	Electronic	East	01-02-2024	2221.104128228201	ARIMA	213	128.39124487004105
S001	P0013	Groceries	North	01-02-2024	1834.4185409544211	ARIMA	356	132.87414500683994
S001	P0014	Groceries	South	01-02-2024	2125.2717757777264	ARIMA	293	136.62106703146375
S001	P0015	Electronic	East	01-02-2024	2543.1653221411734	ARIMA	356	137.88645690834474
S001	P0016	Electronic	East	01-02-2024	4720.501368293754	ARIMA	74	132.9329685362517
S001	P0017	Toys	East	01-02-2024	1284.829291213617	ARIMA	282	136.44870041039673
S001	P0018	Electronic	East	01-02-2024	1596.0977425587855	ARIMA	191	138.81668946648426
S001	P0019	Clothing	West	01-02-2024	2655.0494075233373	ARIMA	149	134.73050615595076
S001	P0020	Electronic	South	01-02-2024	2189.0979639540296	ARIMA	242	138.3638850889193
S002	P0001	Clothing	East	01-02-2024	3202.7096439701213	ARIMA	415	145.00547195622434
S002	P0002	Groceries	North	01-02-2024	1031.5756545914703	ARIMA	364	128.78659370725035
S002	P0003	Clothing	South	01-02-2024	3801.403903152306	ARIMA	372	133.00547195622434
S002	P0004	Toys	West	01-02-2024	2266.893540409774	ARIMA	133	138.4268125854993
S002	P0005	Toys	West	01-02-2024	2450.771751457414	ARIMA	464	132.5718194254446
S002	P0006	Toys	East	01-02-2024	2504.941438136852	ARIMA	160	133.26675786593708
S002	P0007	Toys	West	01-02-2024	5366.110600353066	ARIMA	53	137.62106703146375
S002	P0008	Furniture	North	01-02-2024	2761.114396	ARIMA	218	128.8987688098495
S002	P0009	Electronic	North	01-02-2024	5547.414479629105	ARIMA	62	144.1532147742818
S002	P0010	Toys	West	01-02-2024	4140.618939620442	ARIMA	419	135.44733242134063

- Historical retail sales data was prepared using **Python** for demand forecasting.
- **ARIMA** was used as the time-series forecasting method to generate future demand predictions.
- The forecast output contains information such as **Store ID, Product ID, Category, Region, Forecast Date, Forecasted Demand, Inventory Level and Average Daily Demand**.
- The forecasted demand was incorporated into the Power BI analysis.
- **Inventory Level vs Demand Forecast** allows comparison between available inventory and expected demand.
- Forecasting provides an additional view of future demand based on historical sales information.

1.Sales by Months

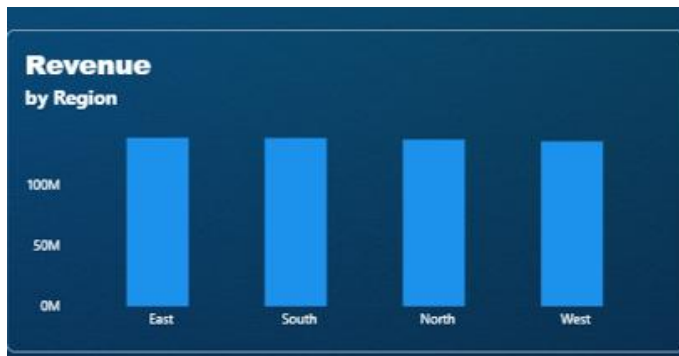


Sales by Months

- The **Monthly Revenue Trend** visual shows the change in total revenue across different months.
- It provides a month-wise view of the retail sales performance.
- The visualization helps identify **increases and decreases in revenue** during different months.
- It makes it easier to observe **monthly sales patterns and variations**.
- This analysis can be used to understand historical sales trends and support **future demand analysis and forecasting**.

Chart Insight: The monthly revenue chart shows noticeable fluctuations in revenue across the year. Revenue increases in some months and decreases in others, indicating variation in monthly sales performance. These patterns can be considered when analyzing seasonality and planning future demand.

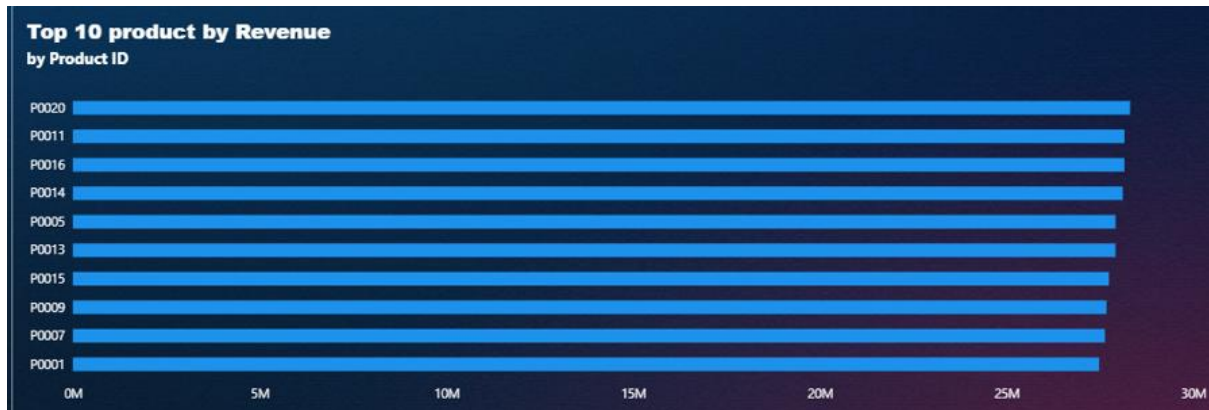
2.Sales by Region



- The **Revenue by Region** visual compares the total revenue generated across different regions.
- The **Units Sold by Region** visual compares the number of units sold across the regions.
- This analysis helps understand the **regional distribution of sales and revenue**.
- It provides a clear comparison of regional sales performance.
- The regional analysis can be used to identify differences in **revenue contribution and sales volume** across regions.

Chart Insight: The regional charts show differences in revenue and units sold across the four regions. The comparison helps identify variations in regional sales contribution and provides a clearer view of regional performance.

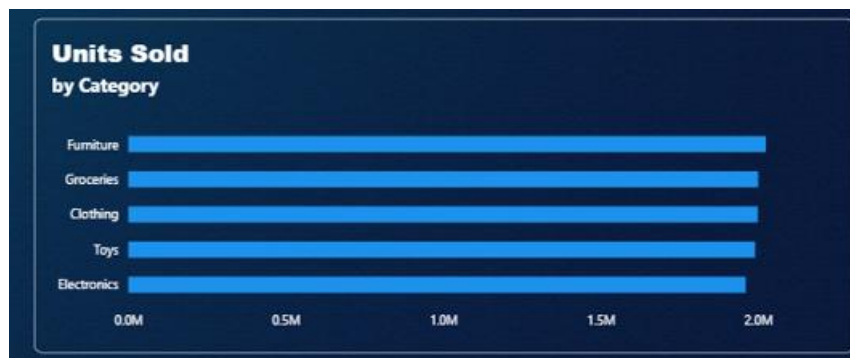
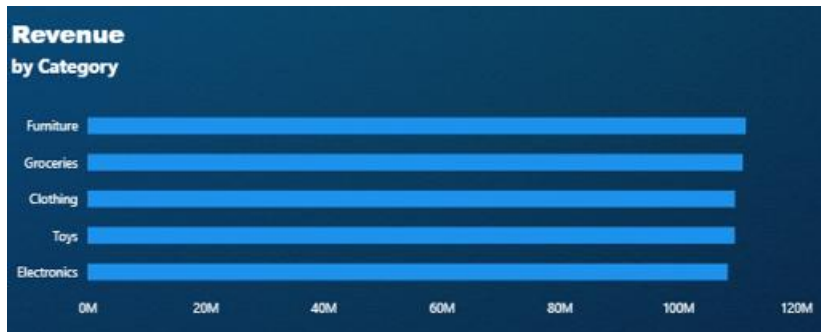
3.Sales by Product



- The **Top 10 Products by Revenue** visual shows the products that generated the highest revenue.
- It provides a comparison of revenue contribution across individual products.
- The analysis helps identify the products with the highest revenue contribution in the dataset.
- It makes it easier to compare product-level sales performance.
- This analysis can be used to understand which products contribute more to the overall revenue.

Chart Insight: The Top 10 Products by Revenue chart shows that revenue contribution varies across individual products. The higher-revenue products make a larger contribution to overall sales, helping identify important products in the dataset.

4.Sales by Category



- The **Revenue by Category** visual compares the total revenue generated by different product categories.
- The **Units Sold by Category** visual compares the number of units sold across the categories.
- This analysis helps understand the **sales contribution of each product category**.
- It provides a clear comparison of **revenue and sales volume** across different categories.
- The category-wise analysis helps identify variations in **category-level sales performance**.

Chart Insight: The category-wise charts show differences in both revenue and units sold. This indicates that product categories contribute differently to overall sales and helps compare category-level performance.

Inventory by Category



- The **Inventory by Category** visual shows the inventory levels across different product categories.
- It provides a comparison of the **available inventory** for each category.
- This analysis helps understand how inventory is **distributed among different categories**.
- It makes it easier to compare **inventory levels across product categories**.
- The category-wise inventory analysis supports understanding of the overall **inventory position** of the business.

Chart Insight: The inventory chart shows that inventory is distributed across all product categories with some variation in stock levels. This provides a clear view of how available inventory is distributed among the categories.

Inventory Level vs Demand Forecast



- The **Inventory Level vs Demand Forecast** visual compares the available inventory with the forecasted demand.
- It helps understand whether the current **inventory level is higher or lower than the expected demand**.
- The comparison provides a clear view of the relationship between **existing inventory and forecasted demand**.
- It can help identify products or categories where **inventory and expected demand differ**.
- This analysis supports better understanding of **inventory position and future demand requirements**.

Chart Insight: The chart shows differences between available inventory and forecasted demand across products. Comparing these values helps identify products where inventory levels may differ from expected demand and supports inventory planning.

Conclusion

The **E-Commerce Inventory & Sales Forecasting** project demonstrates the practical application of **Python, MySQL and Power BI** for retail data analysis. Python was used for **data cleaning, preparation and demand forecasting using the ARIMA method**. MySQL was used to store the retail data and perform SQL-based analysis of **revenue, units sold, categories, regions, monthly sales and products**. Power BI was used to create interactive dashboards for **sales performance, inventory monitoring, product analysis and regional analysis**. The dashboards provide a clear view of **sales trends, category performance, product revenue, regional performance, inventory levels and forecasted demand**. The **Inventory Level vs Demand Forecast** analysis provides an additional view of the relationship between available inventory and expected demand. Overall, the project helped develop practical skills in **data preprocessing, SQL, time-series forecasting, data visualization and dashboard development**, and demonstrated how raw retail data can be transformed into meaningful business insights.

Bibliography

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Author: Anirudh Singh Chauhan

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- **Statsmodels:** Used for ARIMA-based time-series demand forecasting.
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